PAPER 2 1 hour

Essay

Answer **four** *questions* **only**.

All questions carry equal marks.

All working must be clearly shown. Marks will **not** *be awarded for correct answers without corresponding working.*

1. (*a*) Given that $P = \{\text{factors of } 36\}$ and $Q = \{\text{factors of } 54\}$,

(i) list the members in the sets P and Q.

(ii) Find:

(α) $P \cap Q$;

(β) $n(P \cap Q)$;

(γ) the Highest Common Factor (HCF) of 36 and 54.

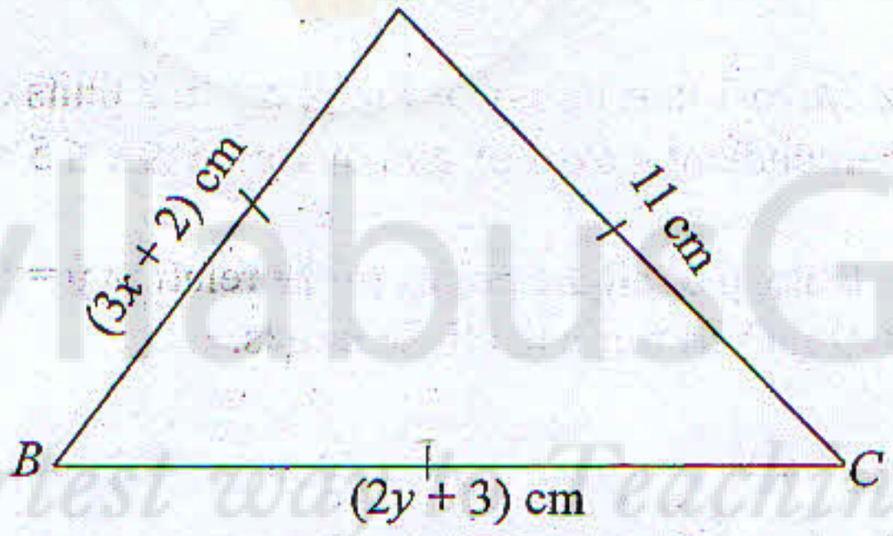
(*b*) Write down the next two terms of the sequence 1, 4, 9, . . ., ...

(*c*) The median of the ordered set of observations 2, 3, $(4m - 3)$, $(3m + 1)$, 11 and 13 in ascending order is 6. Find the value of m.

2. (*a*) Simplify : $\left(\frac{1}{3} + \frac{1}{12}\right) \div \left(\frac{2}{3} - \frac{5}{8}\right)$.

(*b*) Find the product of $(2x - 3)$ and $(2x + 3)$.

(*c*)



NOT DRAWN TO SCALE

In the diagram, ABC is an equilateral triangle. Find the value of $(x + y)$.

3. (*a*) Given the relation $L = \frac{2(m^2 - n^2)}{4(m + n)}$:

(i) simplify L;

(ii) find the value of L when $m = 2$ and $n = 3$.

(*b*) Solve $\frac{4}{3x} = 7 - \frac{3}{x}$.

(*c*) A salesman gets a commission of $5\frac{1}{2}\%$ of the value of items he sells. The salesman sells 12 textbooks at GH¢ 25.00 per book, 3 scientific calculators at GH¢ 50.00 per calculator and 8 packets of bic pens at GH¢ 50.00 per packet. Calculate the salesman's commission.

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4. (*a*) Fred is $(x-1)$ years old now. How old:
   (i) was he 4 years ago?
   (ii) will he be 8 years from now?
   (iii) is he now, if his age in 8 years time will be three times his age 4 years ago?

   (*b*) The perimeter of a rectangular cocoa farm is 497 km. The length of the farm is $2\frac{1}{2}$ times the width. Find the:
   (i) width;
   (ii) length of the farm.

5. (*a*) Factorize: $(x-y)(3m+n)-(x-y)(m-2n)$.

   (*b*) Given that $\mathbf{p}=\begin{pmatrix}2-3x\\5-2y\end{pmatrix}$, $\mathbf{q}=\begin{pmatrix}-1\\5\end{pmatrix}$ and $\mathbf{p}-\mathbf{q}=\begin{pmatrix}6\\8\end{pmatrix}$, find the value of $(x+y)$.

   (*c*) (i) Find the truth set of $\frac{x-1}{2} \leq \frac{1}{2}+x$.

   (ii) Illustrate the answer in (i) on the number line.

6. (*a*) Copy and complete the table for the relation $y=5-2x$ for $-3 \leq x \leq 4$.

x	−3	−2	−1	0	1	2	3	4
y	11			5		1		−3

   (*b*) Using a scale of 2 cm to 1 unit on x-axis and 2 cm to 2 units on the y-axis, draw on a graph sheet two perpendicular axes ox and oy for $-5 \leq x \leq 5$ and $-12 \leq y \leq 12$.

   (*c*) (i) Using the table, plot all the points of the relation $y=5-2x$.
   (ii) Draw a straight line through all the points.

   (*d*) Using the graph, find the:
   (i) value of y when $x=-2.6$;
   (ii) value of x when $y=-2.8$;
   (iii) gradient of the line.

END OF ESSAY TEST

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